

07 May 2019 SEMESTER TEST

MODULE DESCRIPTION	:	Database Systems 3.1
MODULE CODE	:	WRDB301 / WRDV301
FACULTY	:	Science
QUALIFICATION	:	Various
DURATION (IN MINUTES)	:	120
TOTAL MARKS	:	60
PAGES	:	6
EXAMINER	:	1. Dr L. Barnard
MODERATOR	:	1. Ms J. Nel
INSTRUCTIONS	:	Answer all questions Follow instructions as per individual questions Read, and complete, attendance slip details
REQUIREMENTS	:	Clean copy of any edition of original <i>Database Principles</i> texts permitted

DO NOT TURN THE PAGE BEFORE BEING TOLD TO DO SO

1. The form provided in Appendix A is used by Mthimkhulu Village Centre (MVC) to record information on people making use of their vehicle and driver hire services. All the information is manually added on the form. The ideal is that the data is added into a MVC vehicle hire database. Currently, only South African citizens may hire a vehicle. [9]

Use the form provided and develop a conceptual model consisting of only the relational schemas that could be used to effectively store the vehicle, driver and hire data in a database. The contractual information recorded in points 1 – 4 is not recorded in the database.

You need to ensure the following:

- Normalised, 3NF relational schemas, clearly indicating primary and foreign keys
- Use appropriate, atomic attributes
- Vehicle, driver and hire information need to be recorded

Instructions for Question 2:

1. The following SQL queries need to be completed in MS Access.
2. Open SaleDB.accdb from your assessment folder (T:\\tests\\WRDV301\\<your username>).
3. Complete the queries using MS Access.
4. Save your queries, using the question number as query name, e.g. 5a or 7b etc.
5. Once you have completed the queries, copy the following to your MS Word document:
 - a. your SQL code and
 - b. make use of the Snipping Tool to copy a portion of your resultant table (which also shows the number of records in your resultant table – see example in Figure 1).

5a

holder_id	Holder_FName
1	Fiona
2	Zizipho
5	Noel
9	Koos
10	Paseletso
11	Ignatious
12	Corne
13	Zebulon
14	Schalk
17	Omphile
19	Hennie
20	Annamarie

ord: 1 of 52

Figure 1 Sample resultant table

6. DO NOT delete the database.
7. Make sure that you have numbered your questions correctly in your MS Word document.

2. The SaleDB keeps track of the employees, customers and sales made to customers. The following relational schemas are used in the database (**primary keys** are in bold and *foreign keys* in italics): [26]

customer(**cust_code**, cust_fname, cust_lname, cust_street, cust_state, cust_balance)

employee(**emp_num**, emp_fname, emp_lname, emp_hiredate, emp_title, dept_num)

invoice(**inv_num**, inv_date, *cust_code*, inv_total, emp_num)

salary_history(**emp_num**, **sal_from**, sal_end, sal_amount)

The following table describes the attributes in the database:

Customer		Employee	
cust_code	Unique customer code	emp_num	Unique employee number
cust_fname	First name	emp_fname	First name
cust_lname	Last name (surname)	emp_lname	Last name (surname)
cust_street	Street address	emp_hiredate	Date hired
cust_state	State (similar to province)	emp_title	Job title
cust_balance	Outstanding balance	dept_num	Department where employee works
Invoice		Salary_history	
inv_num	Unique invoice number	emp_num	Employee number
inv_date	Date of invoice	sal_from	Date this salary was first paid
cust_code	Customer code	sal_end	Last date this salary was paid
inv_total	Total of invoice	sal_amount	Salary amount
emp_num	Employee number who completed the invoice		

Write the SQL to perform the actions as requested per question.

- Display the starting salary for every employee that has ever worked at the company. You need to display the employee number, employee last name, employee first name, the start date of the first (starting) salary and the starting salary amount. Use an appropriate alias for the starting salary amount. Sort the resulting table in descending order of start date. 7
- Display the customer code, customer first and last name (in the format Surname, First name, e.g. Barnard, Lynette) of all customers that had invoices completed by both employee number 83649 and employee number 83677. Display each customer name once only. Sort the output by customer name. 10
- Determine the total monetary value of the invoices per customer from the customer state of AL as well as the number of invoices each customer has. Use appropriate aliases for the total monetary value of the invoices as well as the total number of invoices. Display the customer code, customer first name, customer last name, total value of the invoices and the total number of invoices. Only display those customers who have at least 5 invoices and a monetary value of total invoices higher than R400. 7
- The company management has decided to change the titles of employees that were known as SPECIALIST to CONSULTANT. Update the employee titles in the employee table as per the management decision. 2

3. Consider the following extract from a transaction log file. [4]

2018-05-17	07:30:00	T10 Commit	
2018-05-17	07:30:05	T2 Read C	
2018-05-17	07:30:09	T1 Read A	
2018-05-17	07:30:15	T4 Read C	
2018-05-17	07:30:18	T3 Read B	
2018-05-17	07:30:20	T2 Commit	
2018-05-17	07:30:24	T1 T3 T4	//checkpoint
2018-05-17	07:30:28	T1 Write C	
2018-05-17	07:30:32	T4 Write B	
2018-05-17	07:30:33	T3 Write A	
2018-05-17	07:30:34	T5 Write C	
2018-05-17	07:30:39	T11 Read A	
2018-05-17	07:30:40	T1 T3 T5	//checkpoint
2018-05-17	07:30:42	T6 Read C	
2018-05-17	07:30:43	T3 Read C	
2018-05-17	07:30:46	T6 Read A	
2018-05-17	07:30:47	T5 Commit	
2018-05-17	07:30:52	T7 Read B	
2018-05-17	07:30:54	T7 Read A	
2018-05-17	07:30:55	T3 Abort	
2018-05-17	07:31:15	T9 Read C	
2018-05-17	07:31:16	T9 Commit	

At time 07:31:17 the system fails. Make use of the entries in the log to explain how and why the log will be used to bring the database back to a consistent state by determining which transactions are to be redone (if any) and which are to be undone (if any) in an immediate update environment.

4. Consider the following sequence of operations (S1), listed in the order it is submitted to the scheduler: [11]

T1: R(V); T3: W(X); T2: R(V); T4: W(V); T3: W(Y); T1 commits; T3: W(V); T4: R(Z); T2 commits; T3 commits

At the start of the operation, T4 already holds an exclusive (W) lock on Y.

The timestamps of the transactions are as follows:

T1=601, T2=602, T3=603, T4 = 604

Make use of the Timestamping concurrency control mechanism, implemented using the Wound / Wait scheme, to show how the sequence is handled.

5. Refer to the relational schema given below (**primary keys** are in bold and *foreign keys* in italics). [10]

customer(**cust_code**, cust_fname, cust_lname, cust_street, cust_state, cust_balance,
cust_dateLastPayment)

payment(**pmt_id**, pmt_date, *cust_code*, pmt_amt, pmt_type)

The basic business rules and operational procedures are given below:

- Cust_Balance represents the amount the customer owes. Cust_Balance is increased with every credit purchase and decreased with every payment

- Each new payment made by the customer updates the date of last payment, Cust_DateLastPayment
- Payment type (Pmt_Type) can either be "CASH" for cash or "CARD" for credit card

Note: Use only the attributes provided.

- a) You need to create the transaction (all the operations in SQL) to update the PAYMENT and CUSTOMER tables for the following transaction (Transaction 102). Your transaction should start with BEGIN TRANSACTION and should end with COMMIT. 4

On 4 May 2019, customer 10010 makes a R100 cash payment. The payment ID for this transaction is 2328.

- b) Use Appendix B, containing a partial transaction log and available in your test folder, to answer this question. Insert the table from Appendix B into your Word document and then complete the partial table. 6

Create a transaction log to track the operations of transaction 102 given above. Use the following "before" values in your log:

TRL ID = 1056 (already entered into table)

TRX NUM = 102 (already entered into table)

Cust_Balance = 345.67

Cust_DateLastPayment = 14-Apr-2019

TOTAL 60

**Once you have completed your test, print your answers. Remember to
*Print on Both Sides (flip pages on the long edge).***



VEHICLE AND DRIVER HIRE INDEMNITY FORM

I, the undersigned (your full name) _____,

South African identification (ID) number _____,

residing at (your full street address) _____

herewith confirm that I hereby sign this Indemnity Form on behalf of myself and all passengers (hereafter referred to as "the Clients") while renting the Mthimkhulu Vehicle from **Mthimkhulu Village Centre** (hereafter referred to as "MVC") from _____ (start date) to _____ (end date).

MVC Vehicle information	MVC Driver information
Type: ☐ Bus ☐ SUV ☐ Family car	MVC staff number:
Registration number:	Name:
VIN number:	Contact number:
Make:	
Model:	

- The Clients hereby confirm that
 - no other person/s will be permitted to drive said vehicle other than the duly appointed **MVC** driver.; and
 - should the vehicle be kept overnight, every effort will be made to ensure the vehicle is securely and safely stored.
- The Clients hereby do not hold the **MVC** legally responsible for any claim whatsoever, which may originate during and/or as result of The Clients usage time, for any loss or damage that may or could be caused by any act(s) and/or oversight by the **MVC**, or by any of the employees, and/or persons acting on instruction of the **MVC**. This indemnification remains valid and in force for the full duration of the rental time.
- The **MVC** and/or its employees will under no circumstances be held accountable for any direct or indirect damage, including but not limited to loss of productivity, loss of income, damage or costs of any nature, which The Client, dependant(s), executors, administrators and/or beneficiaries or any person acting on their behalf, may suffer as result of any action(s) and/or neglect and/or misrepresentation by the **MVC** and/or its employees of officials.
- This reservation is binding on me/my dependants, executors, beneficiaries and/or any person acting on my behalf.

Signed at _____, on this _____ day of _____ 2019

The Client (name in block letters)

The Client Signature

As witnesses:

1. _____

2. _____

MVC representative (name in block letters)

Mthimkhulu Signature

Submit with your answer script.

Student number: _____ **Student First Name:** _____ **Student Surname:** _____

<i>TRL ID</i>	<i>TRX NUM</i>	<i>PREV PTR</i>	<i>NXT PTR</i>	<i>OPERATION</i>	<i>TABLE</i>	<i>ROW ID</i>	<i>ATTRIBUTE</i>	<i>BEFORE VALUE</i>	<i>AFTER VALUE</i>
1056	102								